

AI Content Generator Using Google Gemini API

Detailed Project Documentation for CIT Generative AI Project

Introduction

Generative Artificial Intelligence enables machines to create human-like content. This project uses Python, Streamlit, and Google's Gemini API to generate blog posts, captions, LinkedIn posts, product descriptions, and YouTube video ideas.

Problem Statement

Creating engaging content manually requires time and effort. The project provides an automated AI solution for fast and efficient content generation.

Objectives

- Develop an AI-powered content generator.
- Integrate Gemini API with Python.
- Provide a user-friendly Streamlit interface.
- Generate multiple types of content.

Technologies Used

Python
Streamlit
Google Gemini API
Google Generative AI SDK
Visual Studio Code

Methodology

1. User enters topic.
2. User selects content type.
3. Prompt is generated.
4. Request is sent to Gemini API.
5. AI generates content.
6. Result is displayed to the user.

Features

- Blog Post Generator
- Instagram Caption Generator
- LinkedIn Post Generator
- Facebook Post Generator
- Product Description Generator
- YouTube Video Idea Generator
- Download Generated Content

System Requirements

Hardware Requirements

Component	Requirement
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Processor	Intel Core i3 or above
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RAM	4 GB or higher
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Storage	500 MB Free Space
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Internet	Required
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Advantages

Time saving, increased productivity, easy to use, and creative content generation.

Limitations

Requires internet connection and depends on Gemini API availability.

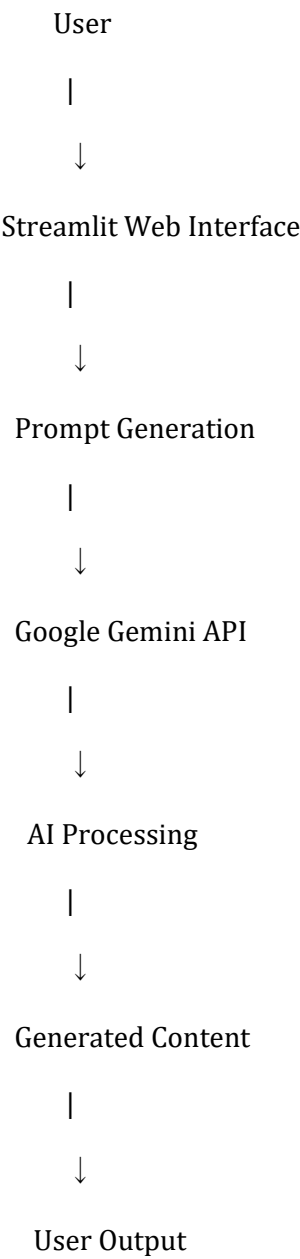
Future Enhancements

Multi-language support, PDF export, dark mode, content history, and image generation.

Conclusion

The project demonstrates the practical implementation of Generative AI for content creation using Python, Streamlit, and Google Gemini API.

System Architechture



FlowChart :

Start

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Enter Topic

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Select Content Type

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Generate Prompt

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Send Request to Gemini API

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Receive AI Response

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Display Generated Content

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End